

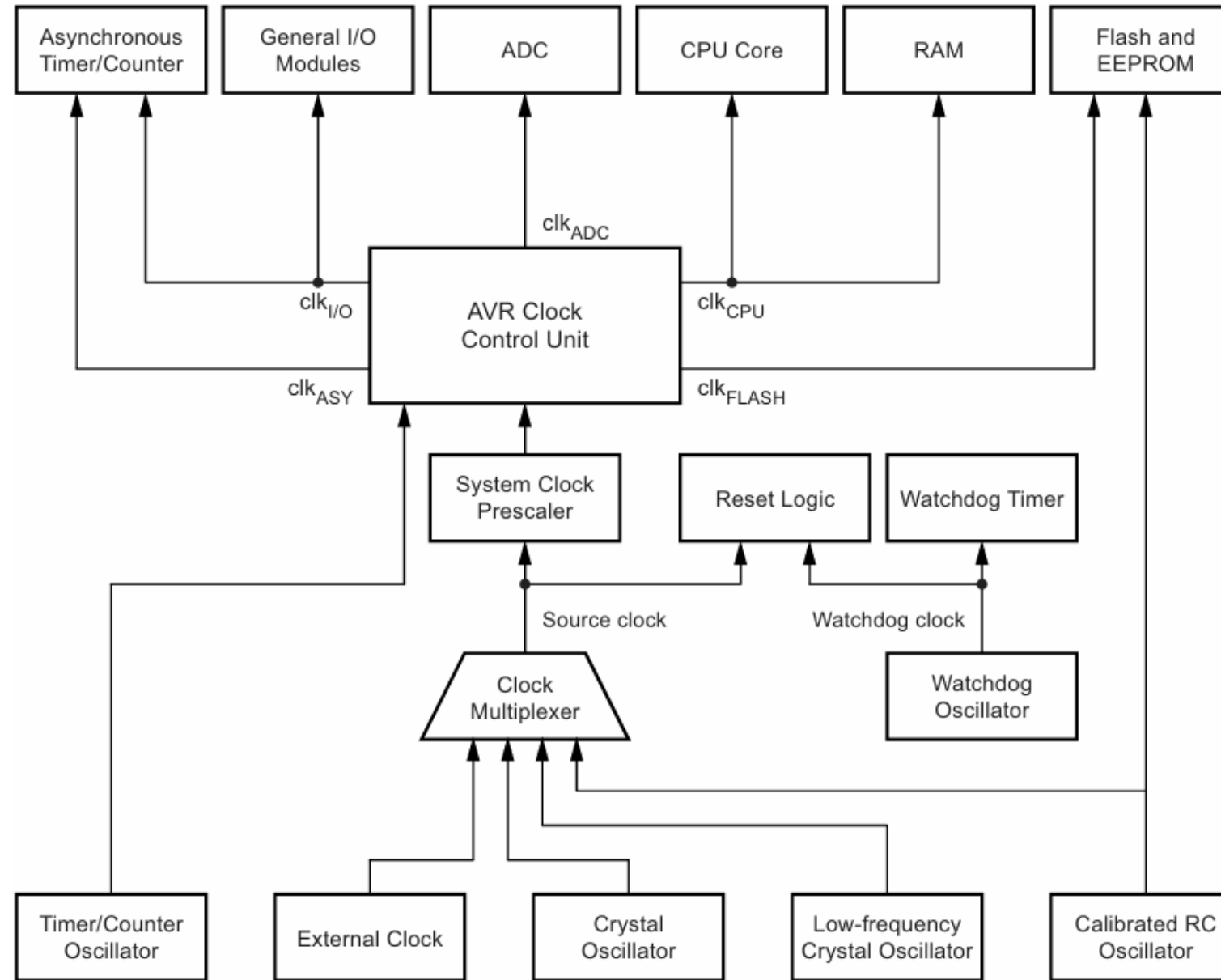
Power Management in ATmega328P

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System Clock and Clock Options



Sleep Mode Control Register

Bit	7	6	5	4	3	2	1	0	
0x33 (0x53)	–	–	–	–	SM2	SM1	SM0	SE	SMCR
Read/Write	R	R	R	R	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- **Bits 7..4 Res: Reserved Bits**

These bits are unused bits in the Atmel® ATmega328P, and will always read as zero.

- **Bits 3..1 – SM2..0: Sleep Mode Select Bits 2, 1, and 0**

- **Bit 0 – SE: Sleep Enable**

The SE bit must be written to logic one to make the MCU enter the sleep mode when the SLEEP instruction is executed. To avoid the MCU entering the sleep mode unless it is the programmer's purpose, it is recommended to write the sleep enable (SE) bit to one just before the execution of the SLEEP instruction and to clear it immediately after waking up.

Active Clock Domains and Wake-up Sources

Table 9-1. Active Clock Domains and Wake-up Sources in the Different Sleep Modes.

Sleep Mode	Active Clock Domains					Oscillators		Wake-up Sources							Software BOD Disable
	clk _{CPU}	clk _{FLASH}	clk _{IO}	clk _{ADC}	clk _{ASY}	Main Clock Source Enabled	Timer Oscillator Enabled	INT1, INT0 and Pin Change	TWI Address Match	Timer2	SPM/EEPROM Ready	ADC	WDT	Other/O	
Idle			X	X	X	X	X ⁽²⁾	X	X	X	X	X	X	X	
ADC noise Reduction				X	X	X	X ⁽²⁾	X ⁽³⁾	X	X ⁽²⁾	X	X	X		
Power-down								X ⁽³⁾	X				X		X
Power-save					X		X ⁽²⁾	X ⁽³⁾	X	X			X		X
Standby ⁽¹⁾						X		X ⁽³⁾	X				X		X
Extended Standby					X ⁽²⁾	X	X ⁽²⁾	X ⁽³⁾	X	X			X		X

- Notes:
1. Only recommended with external crystal or resonator selected as clock source.
 2. If Timer/Counter2 is running in asynchronous mode.
 3. For INT1 and INT0, only level interrupt.

Sleep Mode Selection

Table 9-2. Sleep Mode Select

SM2	SM1	SM0	Sleep Mode
0	0	0	Idle
0	0	1	ADC noise reduction
0	1	0	Power-down
0	1	1	Power-save
1	0	0	Reserved
1	0	1	Reserved
1	1	0	Standby ⁽¹⁾
1	1	1	External standby ⁽¹⁾

Note: 1. Standby mode is only recommended for use with external crystals or resonators.

Power Reduction Register

Bit	7	6	5	4	3	2	1	0	
(0x64)	PRTWI	PRTIM2	PRTIM0	–	PRTIM1	PRSPI	PRUSART0	PRADC	PRR
Read/Write	R/W	R/W	R/W	R	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- **Bit 7 - PRTWI: Power Reduction TWI**

Writing a logic one to this bit shuts down the TWI by stopping the clock to the module. When waking up the TWI again, the TWI should be re initialized to ensure proper operation.

- **Bit 6 - PRTIM2: Power Reduction Timer/Counter2**

Writing a logic one to this bit shuts down the Timer/Counter2 module in synchronous mode (AS2 is 0). When the Timer/Counter2 is enabled, operation will continue like before the shutdown.

- **Bit 5 - PRTIM0: Power Reduction Timer/Counter0**

Writing a logic one to this bit shuts down the Timer/Counter0 module. When the Timer/Counter0 is enabled, operation will continue like before the shutdown.

- **Bit 4 - Res: Reserved bit**

This bit is reserved in Atmel® ATmega328P and will always read as zero.

Power Reduction Register

Bit	7	6	5	4	3	2	1	0	
(0x64)	PRTWI	PRTIM2	PRTIM0	–	PRTIM1	PRSPI	PRUSART0	PRADC	PRR
Read/Write	R/W	R/W	R/W	R	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- **Bit 3 - PRTIM1: Power Reduction Timer/Counter1**

Writing a logic one to this bit shuts down the Timer/Counter1 module. When the Timer/Counter1 is enabled, operation will continue like before the shutdown.

- **Bit 2 - PRSPI: Power Reduction Serial Peripheral Interface**

If using debugWIRE on-chip debug system, this bit should not be written to one.

Writing a logic one to this bit shuts down the serial peripheral interface by stopping the clock to the module. When waking up the SPI again, the SPI should be re initialized to ensure proper operation.

- **Bit 1 - PRUSART0: Power Reduction USART0**

Writing a logic one to this bit shuts down the USART by stopping the clock to the module. When waking up the USART again, the USART should be re initialized to ensure proper operation.

- **Bit 0 - PRADC: Power Reduction ADC**

Writing a logic one to this bit shuts down the ADC. The ADC must be disabled before shut down. The analog comparator cannot use the ADC input MUX when the ADC is shut down.